

Line spectra

Learning outcomes

By the end of this section you should be able to:

- Qualitatively describe the relationship between colour, wavelength, frequency and energy across the electromagnetic spectrum.
- Distinguish between a continuous and a line spectrum.

Elements from the Sun

The element helium (symbol He) was first discovered in 1868 by a French astronomer named Pierre Jules César Janssen. While viewing a solar eclipse through a prism*, he observed a bright yellow line originating from the Sun's chromosphere. The element was named helium after the Greek word 'helios' meaning Sun. The study of the interaction between matter and light is called spectroscopy. Using this technique, elements can be identified by observing their unique emission spectra just like people can be identified by their fingerprints. In this section, we will learn how these emission spectra are produced in terms of electron transitions between energy levels in the atom.

* Be aware that you should never look directly at the Sun; doing so might cause serious eye damage.

Creativity, activity, service

Strand: Service

Learning outcomes:

- Demonstrate how to initiate and plan a CAS experience
- Demonstrate engagement with issues of global significance

The Sun is the source of all of the Earth's natural light. Not only does it provide the necessary light for plants to grow, it can also be used as an energy source in the form of solar power  (<https://www.energy.gov/eere/solar/how-does-solar->

work). One important utilisation of solar power is the solar oven (<https://www.treehugger.com/what-are-solar-ovens-5088602>), a device that uses heat and light from the Sun to cook food. According to Solarcookers.org (<https://www.solarcookers.org/>), approximately one in three people lack sufficient fuel to cook food.

A possible CAS experience could be to raise awareness of the use of solar power, particularly among countries where people lack access to sufficient fuel for cooking.

The electromagnetic spectrum

When you think about light, what first comes to mind? It is probably the visible light that allows us to see. However, visible light is just a small part of what is known as the electromagnetic spectrum, or the EM spectrum (**Figure 2**). The EM spectrum is divided into seven regions arranged in order of frequency, wavelength or energy. The wavelength, λ , is the distance between two crests in an oscillating wave. It has units of distance (m). The frequency, f , is the number of waves that pass a point in one second. It has the units hertz (Hz) or s^{-1} . All regions of the EM spectrum travel at the same speed – the speed of light, which is $3 \times 10^8 \text{ m s}^{-1}$. Frequency and wavelength are inversely proportional, meaning that if frequency increases, the wavelength decreases. **Figure 1** shows two waves – wave A has a longer wavelength and lower frequency than wave B.

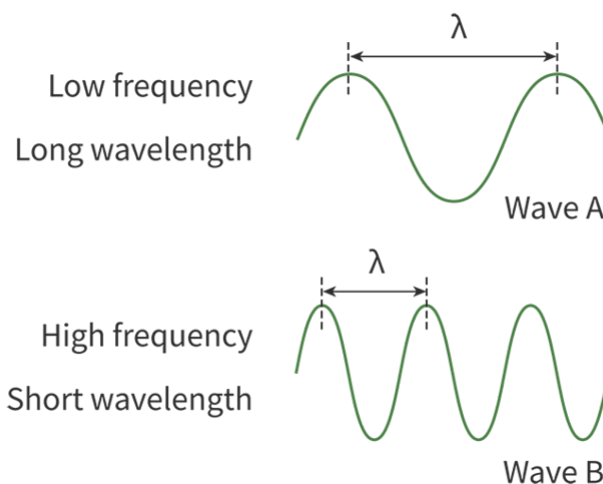


Figure 1. The relationship between wavelength and frequency.

 More information for figure 1

The image shows two distinct wave patterns labeled Wave A and Wave B to illustrate the relationship between wavelength and frequency. Wave A is depicted with a longer wavelength and lower frequency, meaning the distance between consecutive crests is larger, and fewer waves pass a point per second. Wave B, on the other hand, has a

shorter wavelength and higher frequency, indicating a shorter distance between crests with more waves passing a point per second. Both waves are shown in a simple sinusoidal form on separate horizontal lines for clear comparison.

[Generated by AI]

The energy and frequency of electromagnetic radiation are directly proportional, meaning that if the energy increases so does the frequency. High energy corresponds to high frequencies and short wavelengths, and low energy corresponds to low frequencies and long wavelengths.

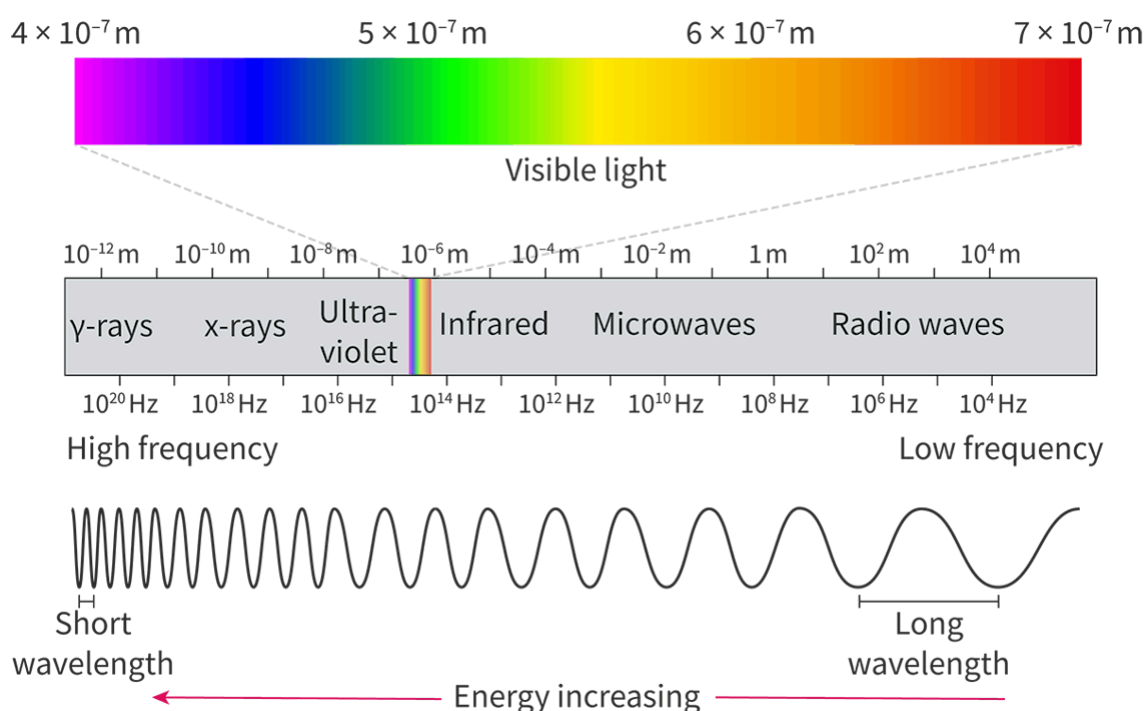


Figure 2. The electromagnetic spectrum.

 More information for figure 2

The image displays the electromagnetic spectrum with various types of waves represented, ranging from gamma rays on the left to radio waves on the right. At the top of the image, a gradient illustrates the change in wavelength and frequency, with short wavelengths and high frequencies on the left, transitioning to long wavelengths and low frequencies on the right. Beneath the gradient, different sections are labeled as gamma rays (γ -rays), X-rays, ultraviolet, visible light, infrared, microwaves, and radio waves. The visible light spectrum is highlighted with a small rainbow-colored band. Text at the bottom indicates areas of high energy/frequency and short wavelengths (on the left) and low energy/frequency and long wavelengths (on the right).

[Generated by AI]

Study skills

The electromagnetic spectrum can be found in [section 5](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-electromagnetic-spectrum-id-45107/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-electromagnetic-spectrum-id-45107/>) of the DP chemistry data booklet.

You should understand the terms directly proportional and inversely proportional. Make sure you are familiar with these terms and are able to use them correctly when explaining the relationship between frequency and wavelength, for example.

The seven regions of the electromagnetic spectrum are shown in **Figure 3**.

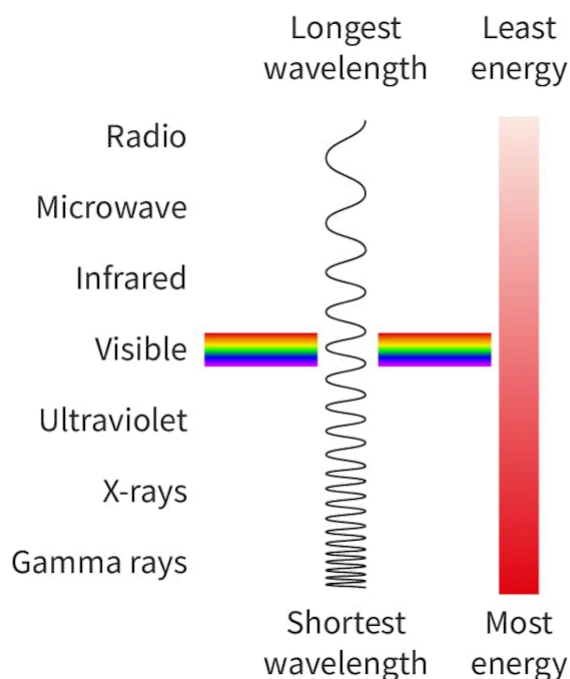


Figure 3. The seven regions of the electromagnetic spectrum.

 More information for figure 3

The diagram illustrates the seven regions of the electromagnetic spectrum, from radio waves to gamma rays. The spectrum is represented vertically with radio waves at the top and gamma rays at the bottom. To the left, labels indicate the types of radiation: Radio, Microwave, Infrared, Visible, Ultraviolet, X-rays, and Gamma rays. In the center, a sinusoidal waveform illustrates the varying wavelength, from longest at the top (radio) to shortest at the bottom (gamma rays). To the right, a gradient bar indicates energy levels, from least energy at the top to most energy at the bottom. Above the waveform,

the label 'Longest wavelength' appears, and below, 'Shortest wavelength.' To the right of the gradient bar, 'Least energy' is labeled at the top, transitioning to 'Most energy' at the bottom. The visible spectrum is highlighted in a rainbow of colors in the middle of the diagram. This visual representation shows how wavelength decreases and energy increases across the electromagnetic spectrum.

[Generated by AI]

Of the seven regions in the electromagnetic spectrum, radio waves have the lowest energy, lowest frequency and the longest wavelength. As we go across the spectrum, the energy and frequency both increase and the wavelength decreases. Therefore, gamma rays have the highest energy, highest frequency and the shortest wavelength.

Figure 4 shows the visible region in more detail, which is known as the continuous spectrum. Our eyes see the continuous spectrum as white light, but as you can see it is actually made up of different colours.

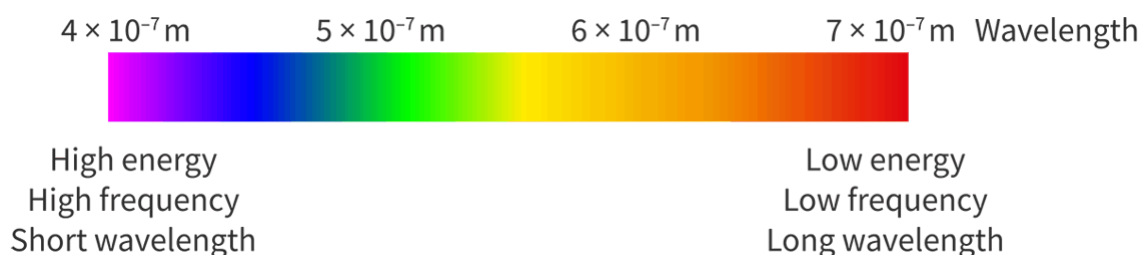


Figure 4. The visible region of the electromagnetic spectrum.

 More information for figure 4

The image shows the visible region of the electromagnetic spectrum as a continuous band of colors transitioning smoothly from violet on the left to red on the right. The colors include violet, blue, green, yellow, orange, and red, representing the range of visible light that our eyes can perceive. The spectrum highlights the concept of white light as a mixture of these various colors, correlating with the description provided before and after the image.

From the text provided after the image, red light is noted to have the lowest energy, frequency, and longest wavelength, whereas violet light, on the opposite end of the spectrum, has the highest energy, frequency, and shortest wavelength. This detailed illustration emphasizes the gradual shift in energy levels and wavelengths across the visible spectrum, making it distinct and informative about the nature of light and color.

[Generated by AI]

In the visible region, red light has the lowest energy, lowest frequency and longest wavelength. As we go across the spectrum, the energy and frequency both increase and the wavelength decreases. So violet light has the highest energy, highest frequency and shortest wavelength.

In **Interactive 1**, drag and drop the seven regions of the spectrum, and the frequency and wavelength, into the correct places.

High energy
frequency
wavelength

Low energy
frequency
wavelength

--	--	--	--	--	--	--

Radio waves

UV

X-rays

Micro waves

γ rays

IR

Low

High

Short

Long

R

G

B

Y

V

O

I

☒ Check

Interactive 1. Electromagnetic spectrum drag and drop.

 More information for interactive 1

This interactive drag-and-drop exercise helps students understand the structure of the electromagnetic (EM) spectrum by engaging them in a drag-and-drop activity. The primary goal is to correctly place the six regions of the EM spectrum, as well as

corresponding labels for frequency, wavelength, and visible light colors, in the appropriate positions based on their relative energies.

At the very top of the interactive activity, you will see two labeled sides. On the left side, above the spectrum, is a label that reads “High energy,” and on the right side is a label that reads “Low energy.”

Directly beneath each of these energy labels are two blank spaces; these are for indicating the frequency and wavelength associated with that energy level. These blanks are meant to be filled in by the user during the drag-and-drop exercise using the provided labels: High, Low, Short, and Long. A zoomed-in view between the third and fourth regions is further divided into seven segments.

Users can drag and drop the six correct regions of the spectrum, arrange the seven segments of the zoomed-in visible light band, and identify whether the frequency is higher or lower and the wavelength is shorter or longer at both the high-energy and low-energy ends of the spectrum.

The draggable gray tiles include;

- Radio waves
- UV
- X-rays
- Microwaves
- λ rays
- IR
- Low
- High
- Short
- Long

Below the main spectrum row is a second, smaller segmented strip with seven blank boxes. This section is for arranging the visible spectrum in the correct order from high to low energy.

Draggable color tiles provided include;

- R (Red)
- O (Orange)
- Y (Yellow)
- G (Green)
- B (Blue)
- I (Indigo)
- V (Violet)

This interactivity helps users understand the electromagnetic spectrum and recognize the properties of waves. At the bottom of the drag-and-drop interactive activity is a "Check" button, designed to allow learners to submit their completed arrangement of the electromagnetic spectrum and receive instant feedback.

Read below for the solution:

The electromagnetic spectrum is divided into six regions as follows:

λ rays

X-rays

UV

IR

Microwaves

Radio waves

In high energy, the frequency is "High" and the wavelength is "Short."

In low energy, the frequency is "Low" and the wavelength is "Long."

The zoomed-in view between the third and fourth regions is divided into seven segments as follows:

V

I

B

G

Y

O

R

This activity provides a powerful visual framework for learners to understand the relationships among energy, frequency, and wavelength across the electromagnetic spectrum.

Emission line spectra

When energy is applied to a sample of hydrogen gas, which is then viewed through a spectroscope, a distinctive pattern can be observed (**Figure 5**).

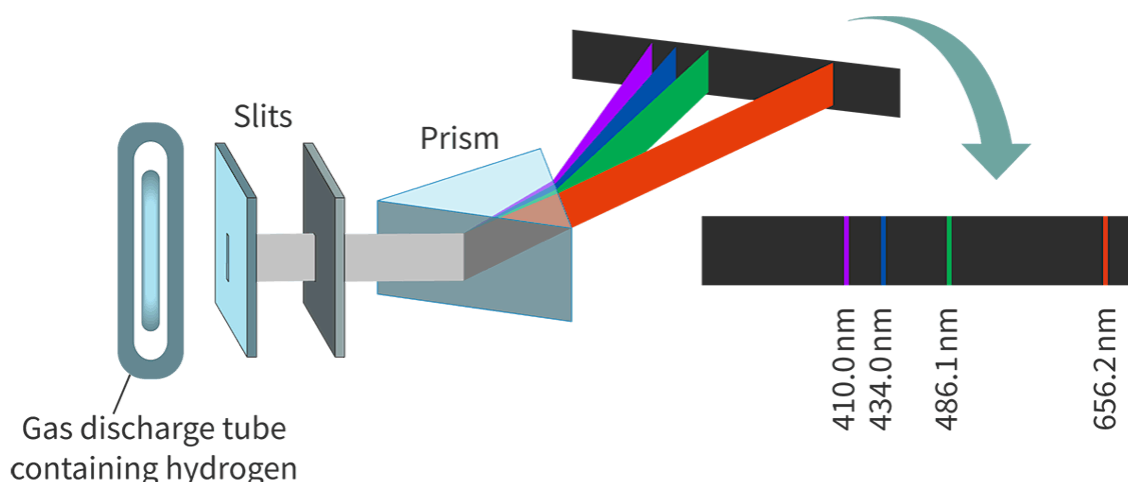


Figure 5. A gas discharge tube containing hydrogen produces a distinctive pattern.

 More information for figure 5

The image is a diagram illustrating the process inside a gas discharge tube containing hydrogen gas. On the left side, there is a tubular structure representing the gas chamber. As energy is applied, light is emitted, passing through various components that seem to include lenses and prisms or diffraction elements. This process results in the dispersion of light into a spectrum. The image shows distinct colored lines representing different wavelengths of light emitted by hydrogen. On the right side, vertical lines in colors such as red, blue, green, and violet represent the emission lines indicating specific wavelengths. These lines correspond to the unique emission line spectrum of hydrogen. The overall structure illustrates how light is collected, dispersed, and viewed, creating a spectrum similar to a fingerprint unique to hydrogen gas.

[Generated by AI]

The pattern observed is known as the emission line spectrum of hydrogen. The emission line spectrum produced is unique to hydrogen, just like fingerprints are to an individual. A comparison between an emission line spectrum and a continuous spectrum is shown in **Figure 6**. What are the differences that you can see between the two?



Figure 6. A comparison of a continuous spectrum (top) and an emission line spectrum (below).

 More information for figure 6

The image displays two types of spectra: a continuous spectrum at the top and an emission line spectrum below. The continuous spectrum shows a smooth gradient of colors ranging from violet to red, representing all visible wavelengths without interruption. In contrast, the emission line spectrum features distinct vertical lines at specific wavelengths against a dark background, corresponding to emission lines of hydrogen. These lines are seen at particular positions in blue, cyan, and red, reflecting the unique spectral fingerprint of hydrogen.

[Generated by AI]

- A continuous spectrum shows all the wavelengths or frequencies of visible light from red to violet.
- An emission line spectrum only shows specific wavelengths or frequencies of light. These are shown as coloured lines on a black background.

Looking at the emission line spectrum of hydrogen, what do you notice about spacing between the lines? Are they evenly spaced, or do they get closer together at one end of the spectrum? How does this observation correspond to what we have learned about the relationship between wavelength and energy in the visible light spectrum?

The lines in the emission spectrum get closer together (converge) at high energy, which corresponds to high frequency and short wavelength. So the distance between the blue and violet lines on the hydrogen emission spectrum is smaller than the distance between the red line and light blue line.

As mentioned before, each element produces its own unique emission line spectrum and these can be used to identify unknown elements. **Figure 7** shows a range of emission line spectra for a selection of elements together with an unknown element. From these can you identify the unknown element?

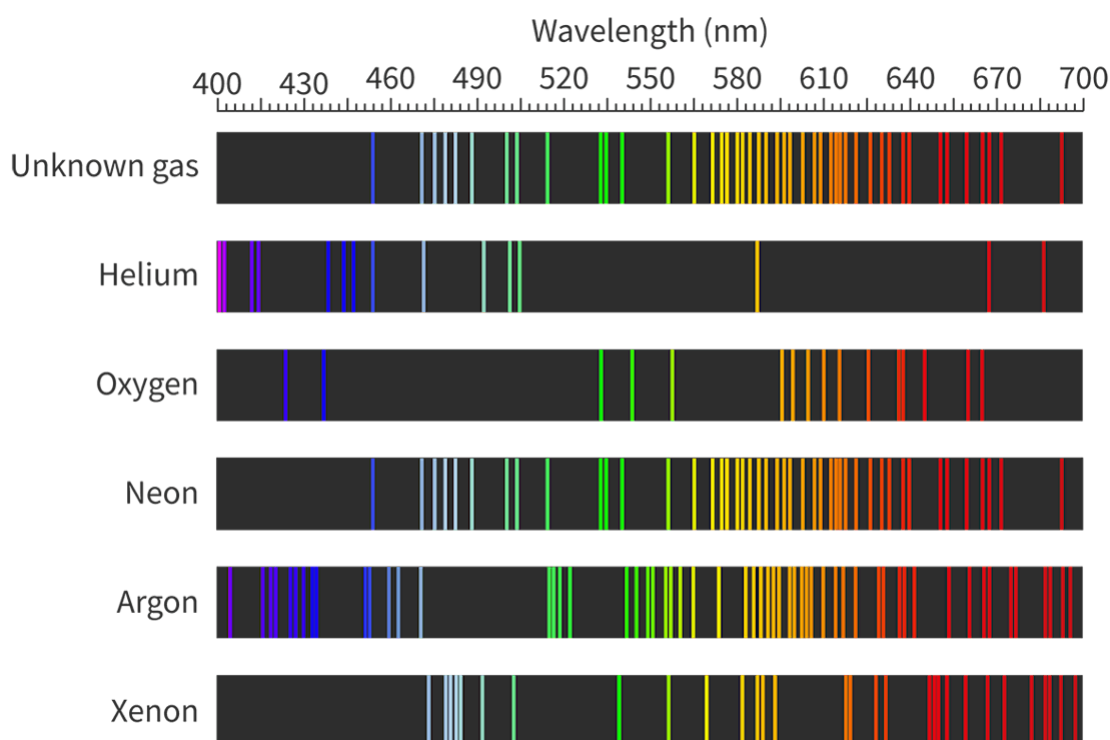


Figure 7. A selection of emission line spectra for certain elements — can you identify the unknown gas?

 More information for figure 7

The image displays several emission line spectra, each consisting of colored lines arranged vertically on a black background. Each spectrum features lines in various positions indicating the spectral emissions of different elements. The colors range from purple and blue on the left to green, yellow, and red on the right. Each group of colored lines represents the unique emission spectrum of a specific element, aiding in the identification of an unknown gas by comparing its emission lines to these known examples.

[Generated by AI]

By comparing the emission line spectra, you can conclude that the unknown gas is neon.

Practical skills

Inquiry 2: Collecting data — Identify and record relevant qualitative observations, Collect and record sufficient relevant quantitative data

In the study of emission spectra from gaseous elements and from light, what qualitative and quantitative data can be collected from instruments such as gas discharge tubes and prisms?

The formation of emission line spectra will be covered in more detail in section S1.3.3-4 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/main-energy-levels-and-sublevels-id-44043/>). For now, just consider which sub-atomic particle and which region of the electromagnetic spectrum is responsible for producing these line spectra?

To understand the formation of an emission spectrum requires us to revisit the model of the atom that was introduced in section S1.2.1a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/atomic-structure-id-43075/>), known as the Bohr model of the atom. Have you ever

wondered why electrons are not stuck to the nucleus? According to Bohr's model, electrons are located in energy levels surrounding the nucleus. **Figure 8** shows the energy levels of an atom (note that the contents of the nucleus are not shown as we are not concerned with them here).

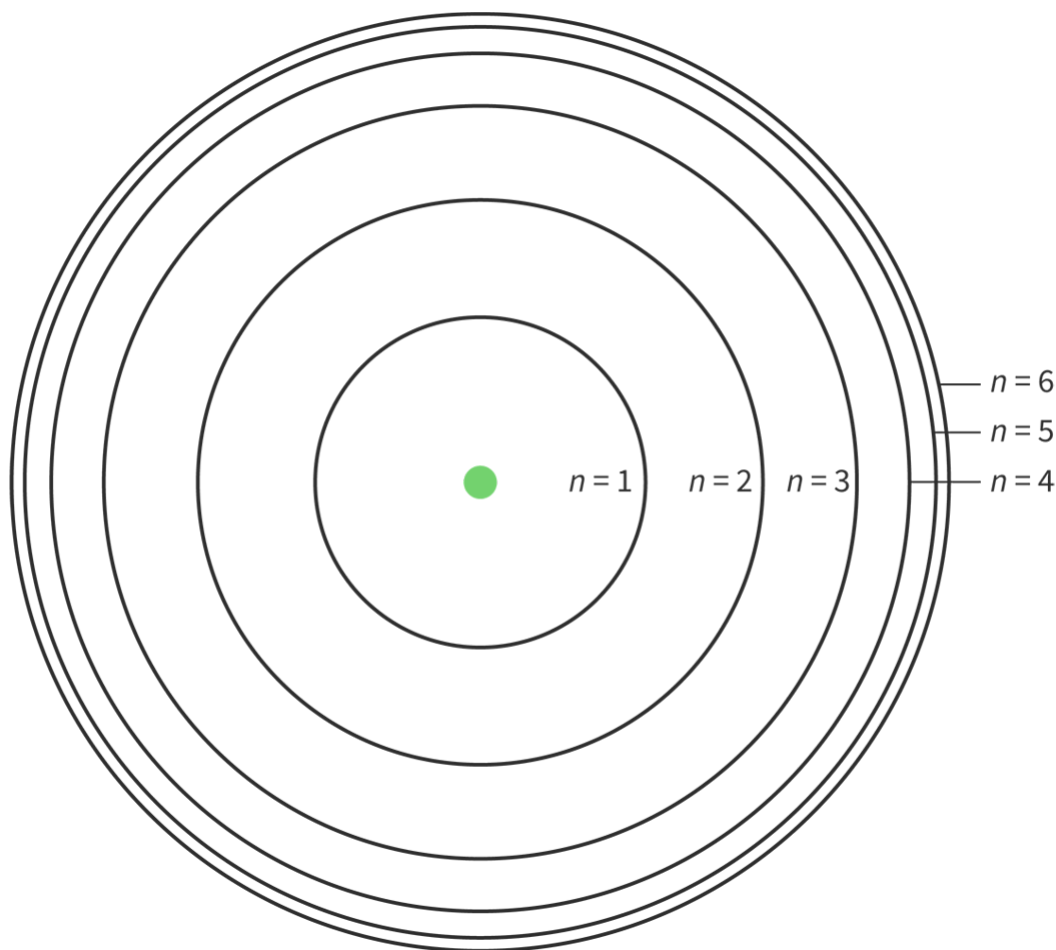


Figure 8. The main energy levels in an atom from $n = 1$ to $n = 6$.

 More information for figure 8

This diagram shows the main energy levels around an atom. It depicts six concentric circles around a central nucleus, representing energy levels from $n=1$ to $n=6$. The first circle closest to the center is labeled $n=1$, indicating the ground state. As the circles extend outward, they are labeled consecutively with $n=2$, $n=3$, $n=4$, $n=5$, and $n=6$, showing increasing energy levels further from the nucleus. The spacing between these circles decreases as they get further from the nucleus, illustrating the convergence of energy levels at higher levels, resembling the behavior observed in an emission spectrum where lines converge at higher energy levels. The diagram emphasizes the principle quantum number ' n ' assigned to each energy level, indicating increased distances and energy states as the value of ' n ' increases. This visualization aligns with the Bohr model's concept of electron energy levels in an atom.

[Generated by AI]

The energy levels are assigned the letter n , which is the principal quantum number. Energy level $n = 1$ is closest to the nucleus (known as the ground state). As the value of n increases, so does the distance from the nucleus and its energy. So $n = 1$ has the lowest energy and $n = 6$ the highest. What do you notice about the spacing of the energy levels? Are they evenly spaced, or do they converge? Where have you seen this trend before? You may remember that the lines in an emission spectrum get closer together at high energy – this is because the main energy levels also converge at high energy.

Making connections

You will learn more about how the energy levels are divided into sublevels in sections S1.3.3-4 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/main-energy-levels-and-sublevels-id-44043/>). Keep this model of the atom in mind or revisit this section to review when you get there.

Electrons can move (or transition) between the energy levels by either absorbing or emitting energy. The energy absorbed or emitted is in the form of photons, which are small packets of energy.

- If an electron absorbs a discrete or an exact amount of energy, it will transition from a lower energy level to a higher energy level, for example from $n = 2$ to $n = 3$.
- The electron is now said to be in an excited state. The excited state is unstable relative to the ground state.
- The unstable electron emits the same amount of energy that it absorbed, and it transitions back down to $n = 2$.
- The amount of energy emitted by the electron in the transition from $n = 3$ to $n = 2$ corresponds to the wavelengths of visible light (in this example, red light).
- A red line will be observed on the emission line spectrum.

The amount of energy emitted also depends on the size of the transition. For example, more energy is emitted when an electron returns to $n = 2$ from $n = 5$ than is emitted when an electron returns to $n = 2$ from $n = 3$. However, both of these transitions fall within the visible region as the energy they emit corresponds to the wavelengths of visible light. These electron transitions will be covered in more detail in the next section, S1.3.2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-hydrogen-emission-spectrum-id-44042/>).

Interactive 2, drag the slider across the screen to see how different transitions relate to different colours in the emission line spectrum.



Interactive 2. Colours of the emission spectrum.

 More information for interactive 2

An interactivity allows users to explore how electronic transitions between energy levels correspond to different colors in the emission spectrum.

As the slider is dragged, different transitions appear, represented by downward arrows between energy levels labeled $n = 6$ to $n = 1$ (ground state). Each transition is visualized as colored lines in the spectrum at the bottom of the graph. The color of the lines and the transition of energy levels are as follows. Violet, $n = 6$ to $n = 2$. Indigo, $n = 5$ to $n = 2$. Blue, $n = 4$ to $n = 2$. Red, $n = 3$ to $n = 2$. Initially, no transitions are displayed, but as the slider is moved from right to left, the first visible transition from $n = 3$ to $n = 2$ appears, followed by additional transitions from higher energy levels to $n = 2$.

As the energy levels increase (from $n = 1$ to $n = 6$), the spacing between them decreases. This means that the energy difference between $n = 4$ and $n = 3$ is smaller than the difference between $n = 3$ and $n = 2$. Consequently, transitions involving lower levels, such as from $n = 3$ to $n = 2$, produce higher-energy photons, resulting in colors toward the blue and violet end of the spectrum. In contrast, transitions between higher levels (like $n = 4$ to $n = 3$) produce lower-energy photons, leading to colors toward the red end of the spectrum. This difference in energy differences explains why spectral lines from higher transitions appear farther to the left in the emission spectrum.

Aspect: Evidence

How do emission spectra provide evidence for the existence of different elements? As you have learned in this section, each element has a distinctive line spectrum just like each person has their own unique set of fingerprints. Throughout history, elements have been discovered from their emission line spectra; helium, discussed earlier, being one such example.

In this activity, you will consolidate your understanding about the relationship between energy, wavelength and frequency of electromagnetic radiation.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Providing a reasoned argument to support conclusions
- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Individual/pair activity

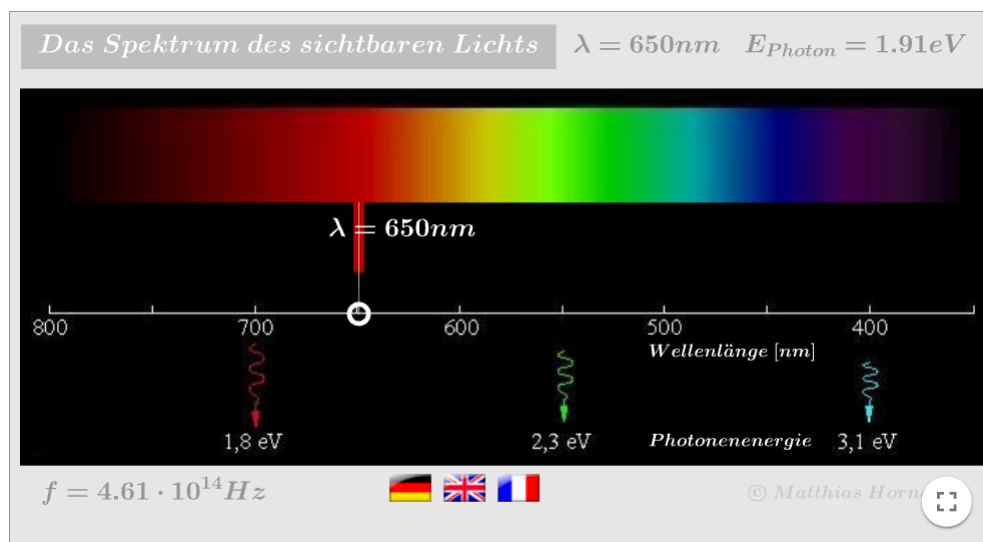
In this activity, you will consolidate your understanding about the relationship between energy, wavelength and frequency of electromagnetic radiation.

Answer the following questions. Once you have finished, compare your answer with a partner.

1. State the relationship between the frequency and wavelength of electromagnetic radiation.
2. State the relationship between the energy and wavelength of electromagnetic radiation.
3. State the relationship between the energy and frequency of electromagnetic radiation.
4. Decide whether the following statements are correct or incorrect. If incorrect, write the correct statement.
 - (a) Gamma rays have the highest energy, highest frequency and longest wavelength.
 - (b) Ultraviolet radiation has a shorter wavelength than X-rays.
 - (c) Microwaves have higher energy and a longer wavelength than infrared radiation.

- (d) Orange light is higher energy than green light.
 (e) Blue light has a shorter wavelength and higher frequency than yellow light.

5. Outline two differences between an emission line spectrum and a continuous spectrum.
 6. Examine the simulation below.



Interactive 3. Energy, Wavelength, and Frequency of Electromagnetic Radiation.

🕒 More information for interactive 3

Move the slider all the way to the left (to the red light). Copy and complete the table below with the wavelength, frequency, energy of red light. Move the slider to the right and record the wavelength, frequency and energy of the remaining colours from orange to violet in the table.

Colour of light	Wave (nm)
Red	
Orange	
Yellow	
Green	
Blue	
Indigo	
Violet	

7. From your table outline the relationship between the wavelength, frequency and energy of the visible region of the electromagnetic spectrum.

1. Frequency and wavelength are inversely proportional.
2. Energy and wavelength are inversely proportional.
3. Energy and frequency are directly proportional.
4. (a) Incorrect, gamma rays have the highest energy, highest frequency and shortest wavelength.
 (b) Incorrect - ultraviolet radiation has a longer wavelength than X-rays.
 (c) Incorrect - microwaves have a lower energy and longer wavelength than infrared radiation.

(d) Incorrect - orange light is lower energy than green light.

(e) Correct.

5. A continuous spectrum shows all the wavelengths or frequencies of visible light whereas an emission line spectrum has only certain wavelengths or frequencies (or colours) shown on a black background.
6. Answers will vary but you should see that as the frequency and energy increase, the wavelength decreases.
7. From red to violet, the energy and frequency both increase and the wavelength decreases.